

First Terminal Examination - 2082

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET - A

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

[11 × 1 = 11]

Rewrite the best alternative of each question in your answer sheet.

- An element A is trivalent and element B is divalent. The formula of the compound formed from these elements will be
 a) A_2B b) AB_2 c) A_2B_3 d) A_3B_2
- Acetylene contains
 a) 1σ and 4π bonds b) 2σ and 3π bonds
 c) 4σ and 1π bonds d) 3σ and 2π bonds
- Which one of the following molecule has highest dipole moment?
 a) H_2O b) H_2S c) HF d) CO_2
- Which of the following molecule is non polar?
 a) HCl b) CH_2Cl_2 c) $CHCl_3$ d) CCl_4
- Which of the following metal ion can be detected by using hydrogen peroxide?
 a) Ca b) Ti c) Fe d) Sc
- Tailing of mercury is due to the formation of:
 a) Mercuric oxide b) Mercurous oxide
 c) Mercuric chloride d) Mercurous chloride
- The false statement is
 a) Isotopes are atoms of different elements
 b) Isotopes are atoms of same elements
 c) Isotopic atoms have same number of protons
 d) Isotopic atoms have same number of electrons
- The volume of 2 moles of a gas at 1 atm pressure and 273K is
 a) 22.4 Litre b) 11.2 Litre c) 44.8 Litre d) 5.6 Litre

9. The molecular formula of oil of vitriol is
 - a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 - b) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
 - c) H_2SO_4
 - d) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
10. Which of the following compounds has two lone pairs and two bond pairs of electrons?
 - a) NH_3
 - b) H_2O
 - c) CO_2
 - d) BF_3
11. In which of the following reactions, H_2 is acting as oxidizing agent?
 - a) $\text{H}_2 + \text{Cl}_2 \rightarrow 2 \text{HCl}$
 - b) $3\text{H}_2 + \text{N}_2 \rightarrow 2 \text{NH}_3$
 - c) $\text{H}_2 + \text{Ca} \rightarrow \text{CaH}_2$
 - d) $\text{H}_2 + \text{CuO} \rightarrow \text{Cu} + \text{H}_2\text{O}$

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. Derive the ideal gas equation $PV = nRT$ where the symbols have their usual meaning. State and explain two conditions under which behaviour of real gas approaches that of an ideal gas. [3+2]
13. Hydrogen is the most abundant element in the universe. The different forms of hydrogen act as reducing agent and it also has three different isotopes.
 - a. List out the isotopes of hydrogen with their number of nucleons. [2]
 - b. Illustrate with two examples to show nascent hydrogen is more powerful reducing agent than molecular hydrogen. [2]
 - c. Mention one use of each of (i) Heavy water (ii) Atomic hydrogen [1]
14. Ammonia, produced industrially by Haber's process and is essential for supplying Nitrogen in Agriculture. It is mainly used in fertilizers, refrigeration and producing various chemicals.
 - a. Write the principle involved in the Haber's synthesis of ammonia with reaction and describe the necessary conditions for the maximum yield of ammonia by this process. [3]
 - b. Draw the flow sheet diagram for the process. [1]
 - c. What is active nitrogen? [1]
15. If sodium metal is reacted with chlorine gas under right conditions, they form a salt called sodium chloride. This is an example of redox reaction.
 - a. What is redox reaction? [1]

- b. Write oxidized and reduced chemical species in the given reaction. [1]
 - c. Identify oxidant and reductant in the given reaction and also define oxidant and reductant with reference to given reaction. [2]
 - d. Calculate the oxidation number of underlined elements
 (i) $\underline{\text{C}}_{12}\text{H}_{22}\text{O}_{11}$ (ii) $\underline{\text{Mn}}\text{O}_4^{--}$ [1]
16. The concept of mole was introduced by Amedeo Avogadro.
 - a. Define relative atomic weight [1]
 - b. Why the atomic weight of element is always fractional? [1]
 - c. Calculate the moles of 102 gram of ammonia gas. [1]
 - d. If the cost of table sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is Rs.100, find its cost per mole. [2]
17. a. Draw the Lewis electron dot structure of [3]
 - i) H_2SO_4 . ii) H_2O and H_2O_2 iii) CO_2 and CO_3^{--}
 - b. How would you explain formation of ionic bond in NaCl and MgCl_2 ? [2]
18. Mutual sharing of equal number of valence electrons between two atoms to attain octet state is called covalent bond.
 - a. What are the conditions of formation of covalent bond? [1]
 - b. Why are the reactions of covalent compounds slower than ionic compounds? [1]
 - c. What is polar covalent compound? Explain with examples. [1]
 - d. What is resonance? Draw the resonating structure of O_3 molecule. [2]
19. The Charle's law can be mathematically expressed as, $V_t = V_0 + V_0 \times \frac{t}{273}$ where V_t and V_0 have their usual meaning.
 - a. Determine the volume of gas at -273°C What is the temperature -273°C called? [1]
 - b. Derive $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ [2]
 - c. Draw graph of V against T based on this law. [1]
 - d. Differentiate Absolute Zero and absolute scale temperature. [1]

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

20. Valence shell electron pair repulsion theory predicts the geometry of a covalent molecule from the number of electron pairs surrounding their central atoms.
- Write all postulates of valence shell electron pair repulsion (VSEPR) theory. [4]
 - Using this theory draw the shape of CH_4 and PCl_5 . [2]
 - Why do NH_3 and BH_3 have different molecular geometries even though their molecular formulas appear similar? [2]
21. Graham's law of diffusion or effusion was formulated by Scottish physical chemist Thomas Graham in 1848. This law describes the relation between density of gas and rate of diffusion.
- State and explain the Graham's law of diffusion with all mathematical expression. [3]
 - How is Graham's law used in everyday life? [1]
 - Calculate the mass of an ideal gas with volume 11200 cc (molecular mass = 17 amu) kept under 750 mm Hg pressure and 27°C temperature. [2]
 - At what temperature, the volume of a certain mass of gas at 30°C will be doubled keeping the pressure constant. [2]
22. Oxygen is the third most abundant element by mass which readily forms oxides with other elements. Some of the oxides are given below

Na_2O	Al_2O_3	CO	SO_2	Fe_3O_4	K_2O_2	N_2O	C_3O_2
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- Identify the acidic oxide, basic oxide, peroxide and mixed oxide from the above table with suitable chemical reactions. [4]
- Write chemical equations to prove that the particular oxide in the above table is amphoteric in nature. [1]
- Why is CO considered as neutral oxide [1]
- Write one industrial and one medical application of oxygen gas. [2]

❧ The End ❧